

# Accuracy as a Fraction

Project: Transparent Book Recommender

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Training and Testing

Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Compare both class fractions for Accuracy and class imbalance. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Accuracy and class imbalance. Proposed explanation: “High accuracy means both classes work well”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

**Independent case: Assess a new comparison independently**

New training labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records. The always-clear baseline is  $[[18,0],[2,0]]$  and candidate is  $[[15,3],[1,1]]$ , with reference rows and clear then blocked on both axes. Calculate ordinary accuracy, both class fractions and balanced accuracy for each. Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

**5. Your learning target**

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two percentages misleading here?

**6. Solve the given case**

Nine of twelve held-out classifications are correct.

Compute accuracy and state the denominator.

Show your trace, calculation or evidence

**Write or draw your reasoning and evidence.**

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## 7. Check a changed case

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 12 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Report correct recommendations / all test users, alongside the raw test count.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two percentages misleading here?

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Score the majority baseline

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, accuracy, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

**My reasoning and evidence**

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## 13. W2 Compare both class fractions

For the worked candidate  $[[12,4],[0,4]]$ , give ordinary accuracy, both class-specific correct fractions, balanced accuracy, false positives and false negatives. Compare with the baseline.

**My reasoning and evidence**

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## 14. W3 Explain a changed class mix

For the illustrative 8-clear 8-blocked set, use baseline  $[[8,0],[8,0]]$  and candidate  $[[6,2],[0,8]]$ . Calculate ordinary and balanced accuracy for each. Explain what stayed fixed and what this example does not prove.

**My reasoning and evidence**

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## 15. W4 Assess a new comparison independently

New training labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records. The always-clear baseline is  $[[18,0],[2,0]]$  and candidate is  $[[15,3],[1,1]]$ , with reference rows and clear then blocked on both axes. Calculate ordinary accuracy, both class fractions and balanced accuracy for each. Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

### My reasoning and evidence

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## 16. W5 Limit claims from missing classes

An evaluation contains ten clear cases and no blocked cases. All predictions are clear. State ordinary accuracy, explain why blocked performance is unavailable, and describe how our code handles the missing class.

### My reasoning and evidence

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## 17. Summaries with counts and proportions — W1 State the denominator

In the worked poll, why is the known denominator 4 rather than 5?

**My reasoning and evidence**

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## 18. Summaries with counts and proportions — W2 Compare proportions

Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare proportions and counts.

**My reasoning and evidence**

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## 19. Summaries with counts and proportions — W3 Repair zero division

All three responses are Unknown. What should the program report?

**My reasoning and evidence**

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## 20. Summaries with counts and proportions — W4

### Summarize independently

Process Yes, Yes, No, Unknown, No, Yes. Report Yes count, known count, unknown count and Yes proportion among known.

#### My reasoning and evidence

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## 21. Summaries with counts and proportions — W5 Avoid overclaiming

Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

#### My reasoning and evidence

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## 22. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 23. Pictorial case practice: Accuracy as a Fraction

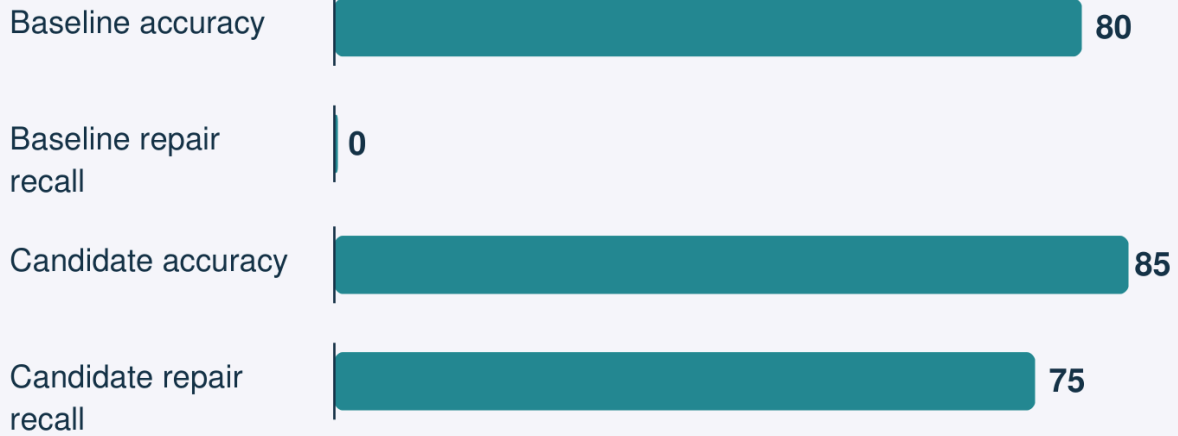
A fictional shelf classifier predicts repair needed or not needed.

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Always-not-needed baseline is correct on 16. Candidate has  $TP=3, FN=1, FP=2, TN=14$ .

## Accuracy can hide missed repair cases

percent



Bar length shows the supplied value; the marked baseline is zero.

## Accuracy as a Fraction

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Reference population

20 labelled cases, including four repair-needed books.



#### 2. Baseline

$16/20=80\%$  accuracy; catches 0/4 needed cases.



#### 3. Candidate

$(3+14)/20=85\%$  accuracy; catches  $3/4=75\%$  needed cases.



#### 4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

### CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Why is 100% repair detection not sufficient to declare the changed classifier best?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate accuracy.